



SCI_LAUNCH_A2_W06 · LAUNCH Science

Autumn 2 Week 6 teacher note

Objective (verbatim): Research a genetic condition online; present.

Scheme of work: LAUNCH scheme of work, Autumn 2 week 6.

Pupil words: I am learning to use a named source to present careful facts about a genetic condition.

Reading gate: LAUNCH 11+; rough FK grade 6.2, age proxy ≈ 12.2 , 995 pupil-visible words excluding marked adult guidance. Formula is a rough screen, not an individual reading assessment.

Pearson foundation route: GCSE Biology 1BI0 Paper 1 and Combined Science 1SC0 Biology Paper 1, shared points 3.12–3.13 (shared Foundation); source evaluation is a teaching approach. Official specification links: <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-biology-spec.pdf> and <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-combinedscience-spec.pdf>. The workbook's teaching approach is not a new specification point. No Biology-only B/HT point is assumed for Combined Foundation.

Gatsby 4: An NHS genetics information worker checks that a patient-facing claim matches the named evidence.

Sequence, success, retrieval

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40 minutes. Three one-minute resets are inside We do, Independent and Review; Arrival and resets make about eight minutes for settling and transitions.

Success at Starter, checked at Exit: I can name the condition and source.; I can explain CFTR inheritance without personal details.; I can tell a sourced fact from a guess.

Retrieval from SoW: “last term” uses the Autumn 1 block in the supplied 2026/27 workbook. If a literal previous academic term is required, staff must check actual teaching rather than invent it.

When	Question	Answer	Exact SoW cell
last lesson	What does a Punnett square show?	Possible allele pairs and their model chance.	LAUNCH scheme of work, Autumn 2 week 5
2-4 weeks	What is a gene?	A section of DNA that codes for a protein.	LAUNCH scheme of work, Autumn 2 week 4
last term	Where are chromosomes found in an animal cell?	In the nucleus.	LAUNCH scheme of work, Autumn 1 week 1

Word meanings and examples (same in lesson and organiser):

Word	Tier	Definition	Example sentence
inherited	3	passed to a child through genes	The card says cystic fibrosis is inherited.
variant	3	a changed version of a gene	The card describes a changed CFTR variant.
carrier	3	a person with one changed copy in this recessive example	A carrier usually does not have cystic fibrosis.
source	2	where a claim comes from	The NHS page is the source for this card.



Faded example: I do full: The NHS card says cystic fibrosis is inherited. Changed CFTR copies come from both parents. One changed copy usually means a carrier; two usually cause the condition. Name the NHS card with each claim. We do part-done: The NHS card says cystic fibrosis is __. It involves the __ gene. One changed copy usually means a __. Source: __. Removed step: Link the two changed CFTR copies to inheritance rather than contact; name the NHS source. Independent uses no filled answer. Fade next: Next lesson hide the “source” sentence and ask pupils to identify the named source independently.

Post-I-do show-me: Show me: is the genetic cause of cystic fibrosis inherited or passed by brief contact? Expected: Inherited. This says nothing about infection precautions a person with CF may need. Every pupil shows a board, point or card. Re-model the named step if insecure.

F/M/H and named stretch

We do F support: Point to the card’s condition, inheritance and care lines before speaking.

We do M: Explain which card sentence supports inheritance and which sentence describes care.

We do H stretch: Use the card to reject the claim that every carrier has symptoms.

Independent F: State the condition and one fact with the NHS source named. Stem: The card says ...; the source is ...

Independent M: Explain CFTR inheritance and separate it from infection. Stem: The CFTR change is inherited because ...; infection advice is separate because ...

Independent H: Revise an unsupported claim with the NHS card. Stem: The claim is not supported by ...; I would revise it to ...

Go deeper: Why does one changed copy in this example not justify saying a carrier has the condition?

Named early-finisher stretch — Source claim audit: Repair a draft saying a CFTR change spreads by touch and outcomes are fixed; source both repairs. This is a harder correction/evaluation, not a repeated item.

On-screen interactions: Match each short statement to a source-card section. uses native selects with a check/reset and aria-live feedback; We do 2 is a native 4-option all-pupil hinge without auto-grading. No storage. Original labelled diagram is simplified, not to scale. Its labels are checked in the fact ledger.

SoW practical status: no core practical is claimed. Prepared model pieces are an optional handling rehearsal with the pupil-facing Risk/Safe lines below.

AfL hinge and misconceptions

Question at We do 2: A fictional family-source card says altered gene copies passed from parents to their child. Which claim does that evidence support?

Option	Correct?	Diagnosis
A. The altered copies spread through a shared cough.	no	Confuses an inherited CFTR variant with an infection spread by a cough.
B. The altered copies arose from the child’s wish.	no	Attributes inherited gene copies to a child’s wish instead of parental transmission.
C. The altered copies were inherited from the parents.	yes	Uses the new case accurately.



Option	Correct?	Diagnosis
D. The altered copies travelled through drinking water.	no	Confuses inherited gene copies with an environmental exposure through water.

Decision: About 3 in 4 or more right → move on. Otherwise re-model an anonymous one-copy/two-copy card and the named NHS line, then ask the parallel hinge. No automatic mark or saved response. Parallel hinge: Another fictional card says altered gene copies were passed through eggs or sperm. Does this support inheritance or person-to-person infection? Answer: inheritance.

Misconception	How to spot	Correction
The genetic cause of cystic fibrosis can spread by contact.	Pupil chooses A, treating the altered copies as contagious.	Use the NHS inheritance line: CFTR changes are inherited. Separately, people with CF may need infection-control advice; do not conflate the genetic condition with infections.
A carrier with one changed copy must have cystic fibrosis.	Pupil chooses C correctly but says one changed copy always means the condition.	Contrast the carrier and two-copy rows in the NHS source card; use “usually” and no real-person claim.

Planned adaptations

Broad area	EEF five-a-day	Concrete action
Cognition & Learning	scaffolding	Keep the NHS source-card model in view; remove one cue for We do, then ask pupils to repeat a fresh example without it.
Communication & Interaction	flexible grouping	Offer the key word card and a private partner turn before speaking to the group; accept point, sign or AAC.
SEMH	cognitive/metacognitive strategies	Check in before the top question and hinge; offer a private first answer, agreed pause and one-step return.
Sensory/Physical	technology	Provide an enlarged high-contrast model and keyboard focus; allow speech-to-text or exact scribing.
EAL (language, not SEN)	explicit instruction	Pre-teach defined science words with examples; let pupils plan in any language while keeping M/H reasoning open.

EAL language support remains available at M and H; the challenge stays open. Speak before write; the organiser's “My language” column is blank, without machine translation. Accept point, sign, speak, draw, type, write, exact scribing and staff-captured photo. Offer speech-to-text, TTS and immersive reader.

TA Focus and prompt ladder

At Independent choose at lesson start: teacher works with supported pupils and TA holds group, or TA supports and teacher holds group. Swap across the week. No public cold-calling is required; this is an invitation bank.

I do: “Watch me underline the exact source sentence for one claim.”

We do: “Which sentence on the card supports your point?”



Independent: “What can you name from the card before adding your reason?”

Least prompt first; no physical prompting: Wait five seconds while pupil finds a source line. → Point to the three labelled parts of the offline card. → Ask “Which line says how the condition is inherited?” → Model underlining one sourced claim; pupil underlines another. → Finish one claim and source together, then pupil tries the next one alone.

Exit handoff: adult states deepest prompt level, valid response mode and what pupil did independently; other adult chooses the next prompt.

Route	Invitation	Expected answer
F	What is the condition named on our card?	Cystic fibrosis.
F	Which organisation made the source page?	The NHS.
M	Why is cystic fibrosis inherited?	Changed CFTR copies are inherited from parents.
M	What does carrier mean here?	One changed CFTR copy, usually without the condition.
H	Which claim about the genetic condition is unsupported: inherited or caught by contact?	Catching the genetic condition is incorrect; infection precautions for people with CF are separate.
H	Why cite the source beside a claim?	So another person can check it and its limits.

Feedback, EHCP broad areas and regulation

Codes exactly: VF · WS · I · NS(+) · E · R · // · ? . VF at We do or Review; WS when supported with recorded prompt level; I when independent at Independent/Exit; NS(+) precise next step at Exit; // a real self-edit at Review; ? adult reads the changed words back on request; R after the pupil responds to NS(+) next lesson and staff judge, never automatic; E only for an actual curriculum observation.

What I record and where: note hinge response split and the named misconception on the staff quick-record grid; record actual code and prompt level in feedback. E: evidence captured on the school's evidence app. Never gather evidence during distress or store pupil data in HTML.

Pathways: high support, VF/WS regular. Build: WS/I mix. Grow: mostly independent, self-edit and response. Launch: independent self-assessment.

Regulation: Check in before the hinge, top rung and Independent. At the agreed pause signal give space and a single return step. Collect no evidence while a pupil is distressed. Re-enter at the last completed step, with one small choice and private correction.

lesson record poster: LAUNCH_SCI_A2_W06. Staff device or projector only; pupil phones not required.

Lundy Exit: pupil can say, point, sign, draw or write; teacher hears the answer and changes the next model/prompt. Four answers: goal at top, actual Review feedback, Exit NS(+), success criteria at Starter and Exit.

Sources, checks and limits

Every scientific and diagram label claim is listed in LESSON_SOURCE.json with a named official source and status. False distractors are deliberately incorrect and diagnosed. Read source claims in context; no medical claim about an individual. The workbook says “online”; the accepted lesson uses a curated offline NHS source card to meet the zero-external-load rule. No pupil research on their own phone and no individual diagnosis. The genetic cause is inherited; people with CF may separately need infection-control advice, as the NHS page explains.



Prepared model handling · practical safety

Kit: Large prepared NHS source cards and claim cards, desk mat

Method: Match the prepared source cards to the claims and cite the named source.

Pupil Risk: Loose source cards could fall onto the floor.

Pupil Safe: Keep cards on the desk mat and pick up any that fall.

Staff confirm the local school risk assessment and exact kit; this does not assert a SoW core practical.

SLT planning and evaluation

Lesson code: SCI-LAUNCH-A2-W06

SoW reference: LAUNCH scheme of work, Autumn 2 week 6

Exact SoW outcome: Research a genetic condition online; present.

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Evidence in this lesson: Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A2-W06. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Look at this week's NHS source card. Point to one detail you notice. Staff: Offer a private, predictable start. Do not require previous knowledge.

Starter (3 min): Recall three ideas from earlier lessons. Show one answer at a time. Staff: All pupils respond; reveal the answers only after attempts. Revisit success criteria.

I Do (5 min): Watch the full model. Point to the step that answers today's question. Staff: Model one complete example; check every pupil's response to the show-me question.

We Do (7 min): Complete the missing step in the part-done model. Speak, then match the cards. Staff: Wait five seconds, gesture, then cue if needed. Support the route without supplying the missing answer.

We Do 2 (5 min): Choose one hinge answer. Show it at the same time as the group. Staff: Collect simultaneous responses, apply the decision rule and parallel hinge in the staff note.

Independent (8 min): Present three sourced facts from the offline NHS card, then correct an unsupported claim. Staff: At lesson start choose one adult with supported pupils and one holding the group; swap during the week. Record independence and prompt level.

Review (4 min): Tell a partner what you changed after feedback. Mark a real edit // and read it back when asked ?. Staff: Give a precise VF or NS(+) only when it actually happens. Invite a true self-edit and read-back.

Exit (3 min): Show what you know with a model and explain one reason. Staff: Hear the answer privately in a valid mode and use it to change the next example or prompt.

Planned adaptation — cognition learning: Keep the NHS source-card model in view; remove one cue for We do, then ask pupils to repeat a fresh example without it.

Planned adaptation — communication interaction: Offer the key word card and a private partner turn before speaking to the group; accept point, sign or AAC.



Planned adaptation — semh: Check in before the top question and hinge; offer a private first answer, agreed pause and one-step return.

Planned adaptation — sensory physical: Provide an enlarged high-contrast model and keyboard focus; allow speech-to-text or exact scribing.

Planned adaptation — eal: Pre-teach defined science words with examples; let pupils plan in any language while keeping M/H reasoning open.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: flexible grouping; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Present three sourced facts from the offline NHS card, then correct an unsupported claim. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W07. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W07: Topics 2 to 3 assessment (Dec).

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Link the two changed CFTR copies to inheritance rather than contact; name the NHS source. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "A fictional family-source card says altered gene copies passed from parents to their child. Which claim does that evidence support?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Look at this week's NHS source card. Point to one detail you notice. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Present three sourced facts from the offline NHS card, then correct an unsupported claim. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use inherited, variant, carrier, source with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.



Cross notation: The simplified CFTR model uses one letter, F/f, with F as the usual copy and f as the changed copy; possible combinations are FF, Ff, fF and ff. This is a model probability for each pregnancy, not a prediction about a person.